

Math Made Clear

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About the Author

Short author bio here. (e.g., background and inspiration; include website/contact if desired.) Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Acknowledgements

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Preface

Mathematics is a universal language, but learning it in a second language can be both a challenge and an opportunity. This book was created especially for Korean students who are studying math in English. Our purpose is to bridge the gap between two worlds—helping you master essential math concepts while also building your confidence in English.

The motivation for this book comes from the experiences of many students who have struggled to understand math vocabulary, instructions, and problem-solving strategies in English. We know that with the right support, every student can succeed. By providing clear explanations, bilingual vocabulary, and step-by-step examples, we hope to make math accessible and enjoyable for you.

Whether you are preparing for international exams, planning to study abroad, or simply want to strengthen your math and language skills, this book is here to support your journey. Remember, every great mathematician started with curiosity and perseverance. We hope this book inspires you to explore, ask questions, and discover the beauty of mathematics in both Korean and English.

Introduction

Welcome to this math book designed especially for Korean students learning English! This book is organized to help you build a strong foundation in mathematics while improving your English skills. Each section covers a key topic, starting with clear explanations and followed by examples, practice questions, and helpful tips.

The book is divided into major sections such as Numbers & Operations, Expressions & Equations, Geometry, Data & Statistics, and more. Within each section, you will find step-by-step guides, vocabulary lists, and real-world applications. Special features include bilingual glossaries, common mistakes to avoid, and mini quizzes to check your understanding.

To get the most out of this book, we recommend reading each section carefully, trying the examples on your own, and using the vocabulary lists to strengthen your math English. Do not hesitate to revisit earlier sections if you need a refresher. Remember, learning math in a new language takes time and practice—be patient with

yourself and celebrate your progress!

We hope this book becomes a valuable companion on your journey to mastering math and English together.

Chapter 1

Numbers & Operations

In this chapter, you will explore the world of numbers and how we use them every day. You will learn about place value, decimals, fractions, and more. Each section includes explanations, examples, and practice problems to help you master these important math skills.

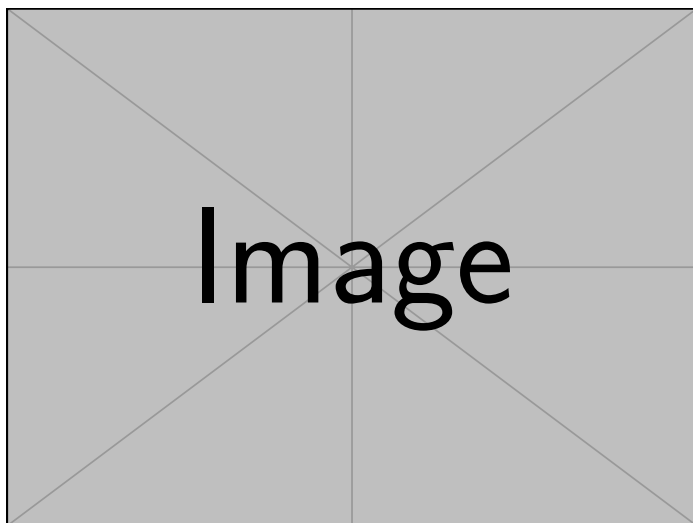


Figure 1.1: A scenic travel destination (example image).

1.1 Place Value

In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number. This idea is called place value. Each place in a number is ten times the value of the one to its right. A number like 4275 is not just four digits. It is a structure: $4000 + 200 + 70 + 5$. We do not read numbers one digit at a time. We read them as a system, where each part works because of its position.

Place value turns a few digits into large or small numbers. Take 9836. The 8 is not just an 8. It is 800, because it sits in hundreds of places. In 5104, the 0 adds nothing but still matters. It shows that there are no tens. In 12345, the 2 sits in the thousands place and means 2000. These examples may seem simple, but they show a key truth: digits mean more when you know where they stand.

Place value is more than vocabulary. It affects how we add, subtract, estimate, and think about numbers. When we know what each digit means, we can break numbers apart and put them back together. We stop guessing and start understanding. That shift helps us see math not as a list of rules but as a system that makes sense.

Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.

Example: Understanding Place Value

To truly grasp the concept of place value, let's take a closer look at the number 4275. Here, the digit 4 is found in the thousands place, which means it actually represents 4000, not just 4. Moving to the right, the 2 is in the hundreds place, so it stands for 200. The 7, sitting in the tens place, is worth 70, and finally, the 5 in the ones place simply means 5. This system, where each digit's value increases by a factor of ten as you move left, is what makes our decimal system so powerful and flexible.

To help visualize this, consider the following place value chart:

Place	Value
Thousands	1000
Hundreds	100
Tens	10
Ones	1

Let's see how this works in practice. If we write 4275 in expanded form, we get:

$$4275 = 4000 + 200 + 70 + 5$$

Now, imagine you are looking at the number 9836. You might wonder, what is the value of the digit 8 in this number? Since the 8 is in the hundreds place, its value is 800. This simple observation shows how the position of a digit determines its true value. Similarly, if you break down 5104 into its expanded form, you see it as $5000 + 100 + 0 + 4$, making it clear what each digit stands for. In another example, the digit 2 in 12345 is in the thousands place, so it represents 2000. Understanding these positions helps us see the true value of every digit in a number, which is a fundamental skill in mathematics.

1.1.1 Hand-drawing1

This is where you will see hand-drawing answer of the same question before in **English**.

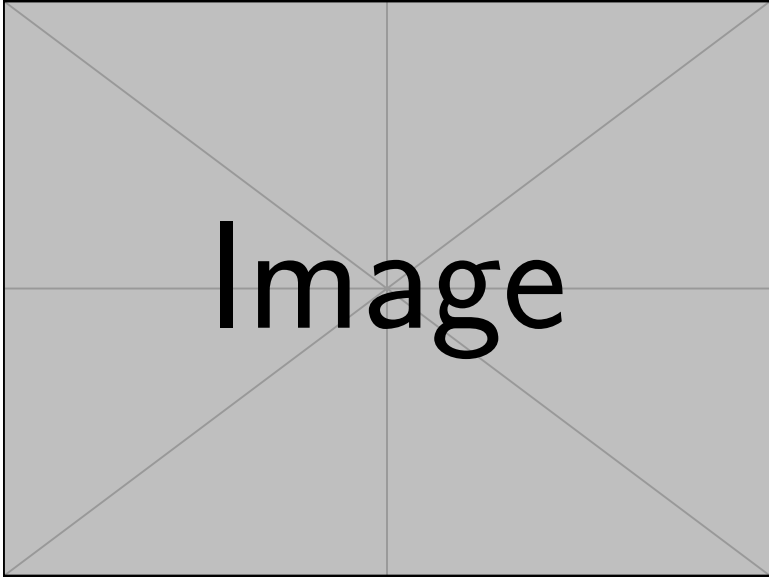


Figure 1.2: A scenic travel destination (example image).

1.1.2 Hand-drawing2

This is where you will see hand-drawing answer of the same question before in **Korean**.

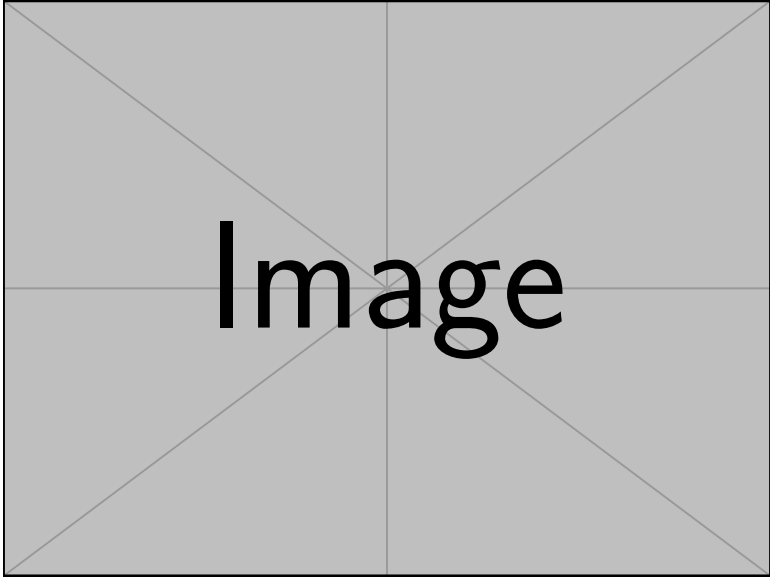


Figure 1.3: A scenic travel destination (example image).

1.2 Concept2

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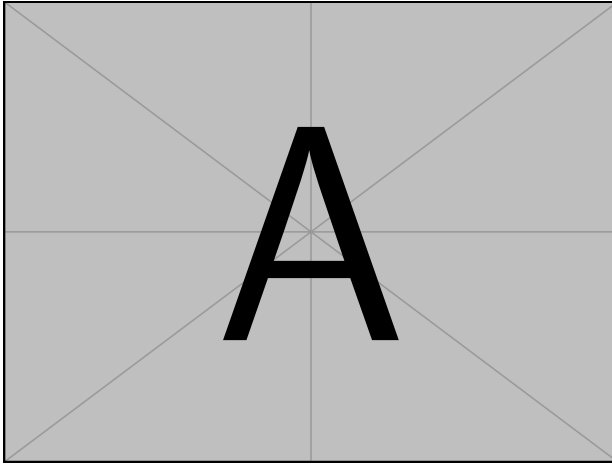


Figure 1.4: Another illustrative image.

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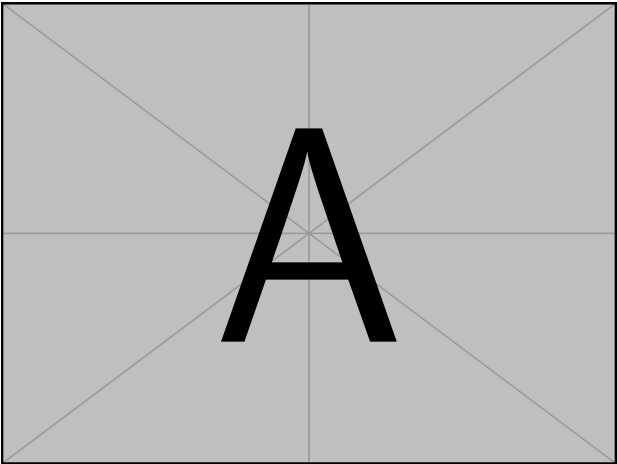


Figure 1.5: Another illustrative image.

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1.2.1 Hand-drawing1

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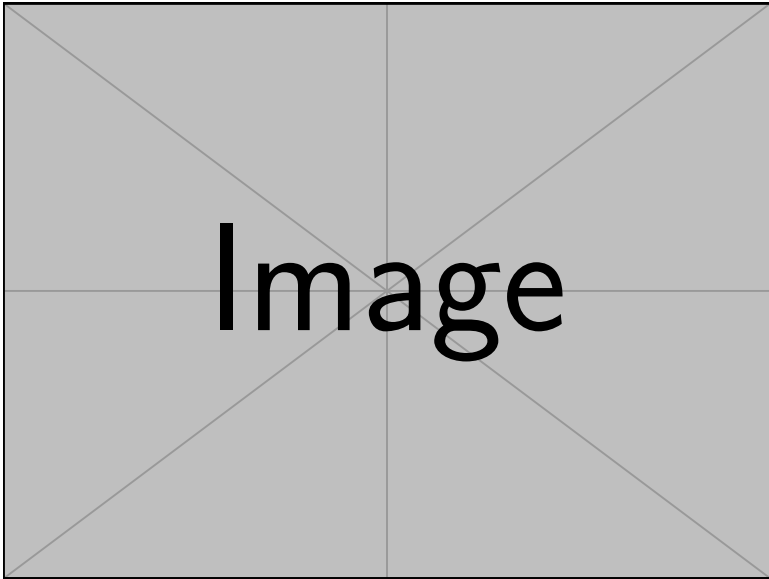


Figure 1.6: A scenic travel destination (example image).

1.2.2 Hand-drawing2

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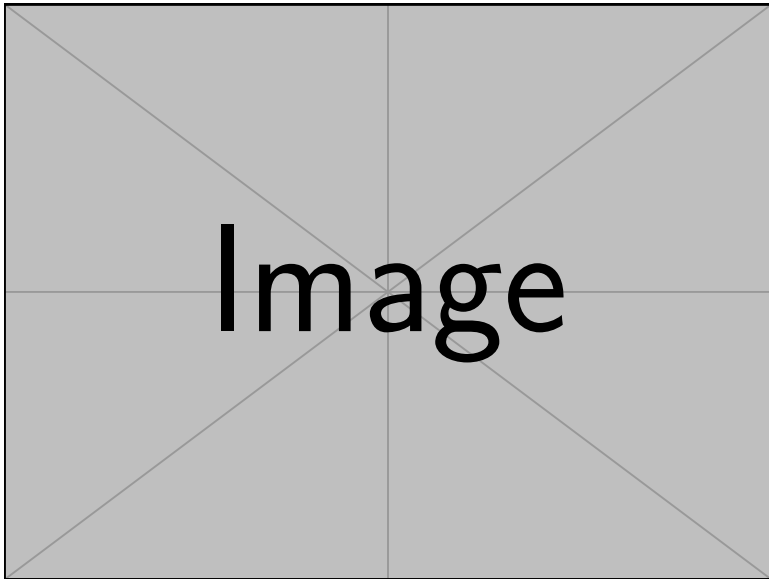


Figure 1.7: A scenic travel destination (example image).

1.3 Concept3

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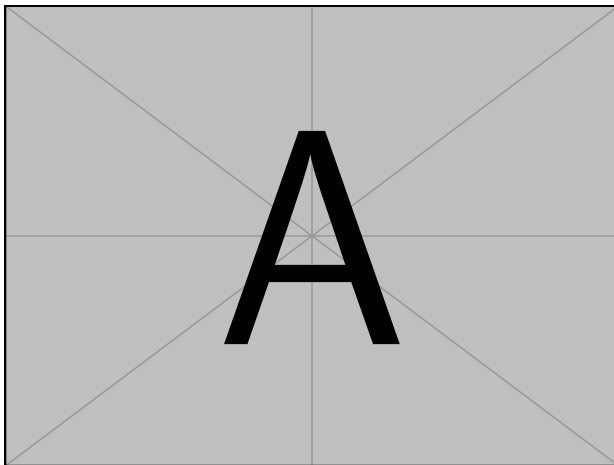


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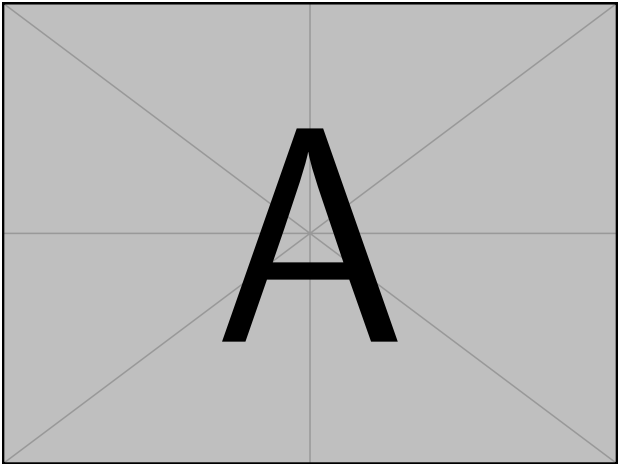


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1.3.1 Hand-drawing1

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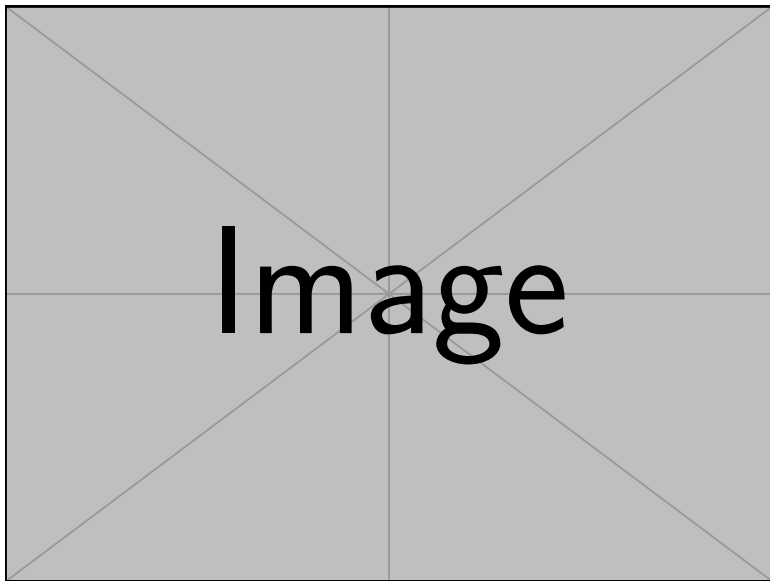


Figure 1.10: A scenic travel destination (example image).

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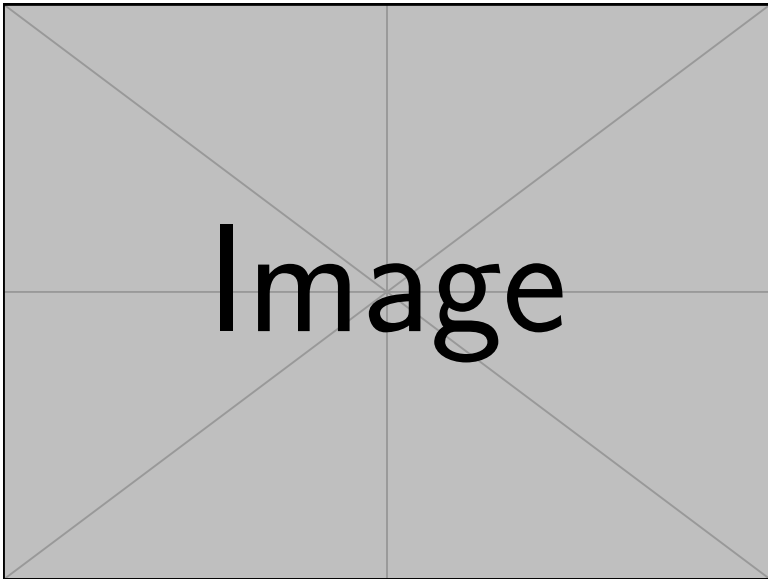


Figure 1.11: A scenic travel destination (example image).

Appendix

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